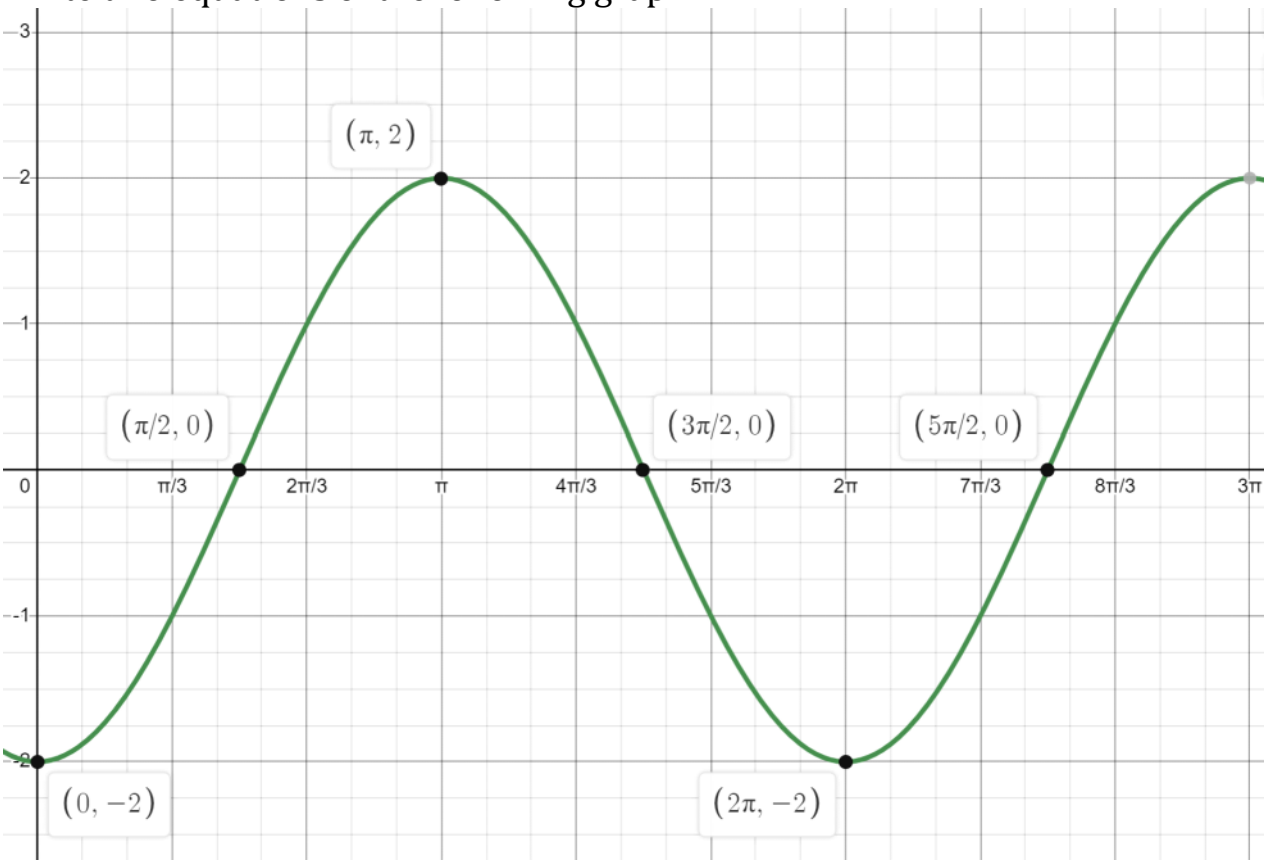
Day 25 HW

Tuesday, March 18, 2025 9:59 AM

Equations from Graphs	2 Problems
Applications	2 Problems

Non-CPM Problem #1:

Write two equations of the following graph:

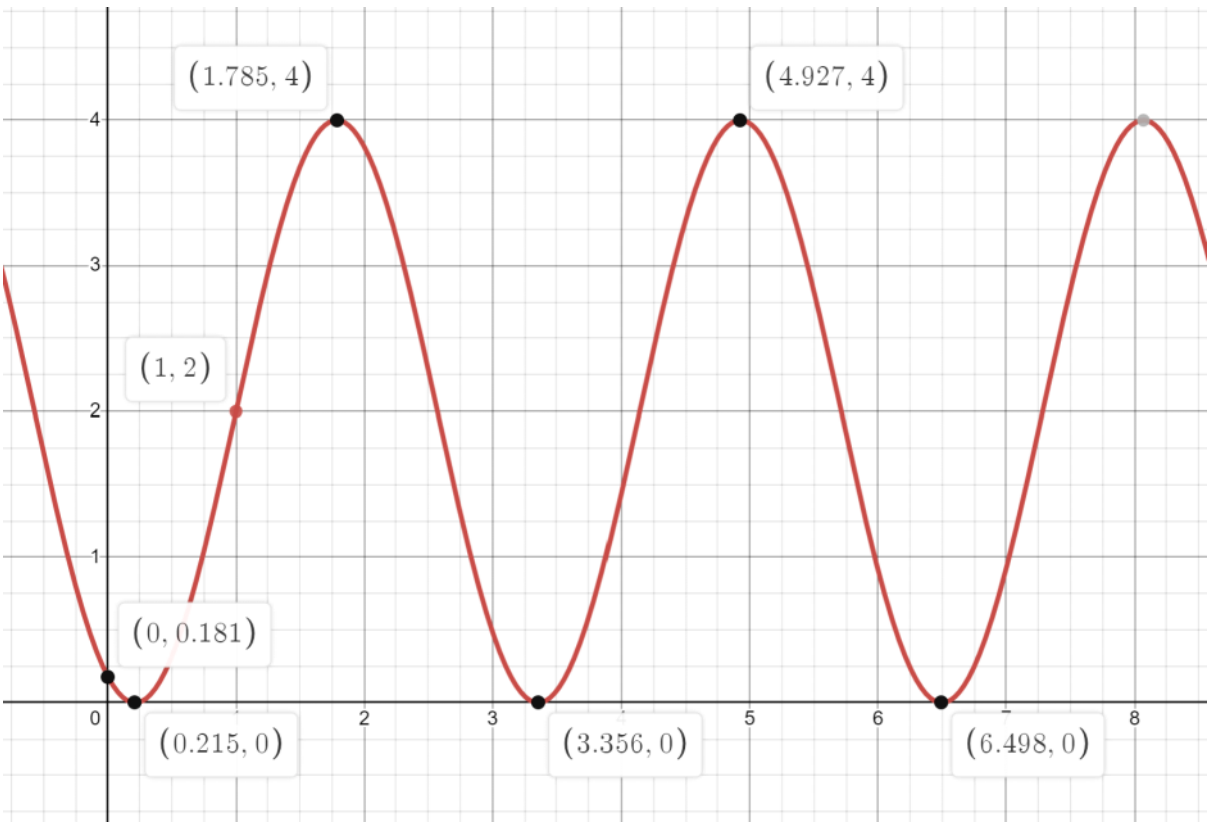


MAKE SURE TO JUSTIFY HOW YOU KNOW

Sine	*Cosine*
Amplitude, a:	Amplitude, a:
Midline, k:	Midline, k:
Horizontal Translation (Phase Shift), h:	Horizontal Translation (Phase Shift), h:

Period, p: Frequency, b: Equation:	Period, p: Frequency, b: Equation:

Non-CPM Problem #2:
Write two equations of the following graph:



MAKE SURE TO JUSTIFY HOW YOU KNOW

Sine Amplitude, a:	*Cosine* Amplitude, a:

Midline, k: Horizontal Translation (Phase Shift), h: Period, p: Frequency, b: Equation:	Midline, k: Horizontal Translation (Phase Shift), h: Period, p: Frequency, b: Equation:

Non-CPM Problem #3: Ferris Wheel Application

A Ferris wheel 120 feet in diameter completes a revolution, rotating counterclockwise, every 40 seconds. The lowest point is 10 feet below ground.

a. Draw and label the Ferris wheel.

a. Draw the graph of the situation, starting with a person getting on at the 3 o clock position at time $x = 0$ seconds.

a. Determine an equation to represent the rider's height above the ground. Be sure to keep your equations in radians.

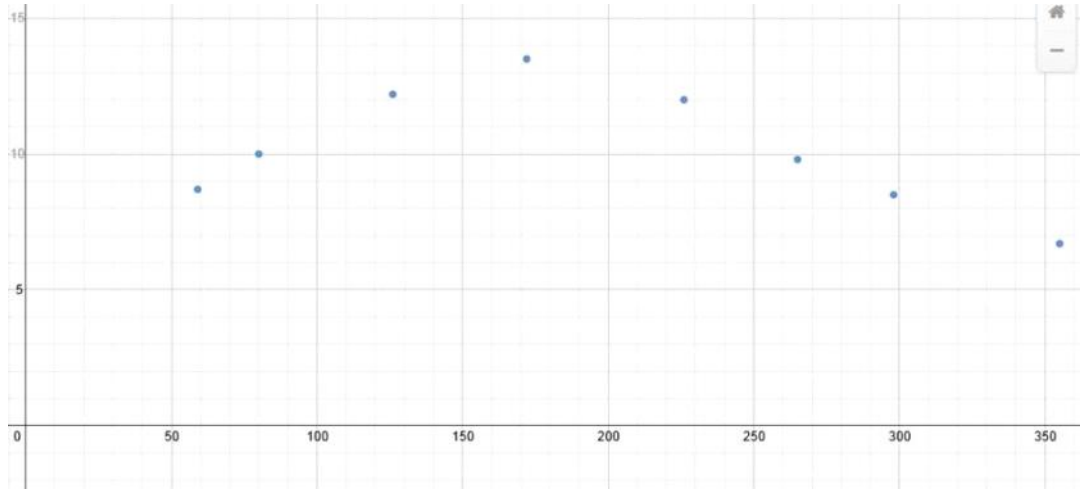
MAKE SURE TO JUSTIFY HOW YOU KNOW

Non-CPM Problem #3: Application

The following data gives the number of hours, y, of night time for Sydney, Australia, as a function of x, the day of the year, starting with January 1 as 1.

a) Sketch the graph to fit the data points.

x	y
59	8.7
80	10
126	12.2
172	13.5
226	12
265	9.8
298	8.5
355	6.7



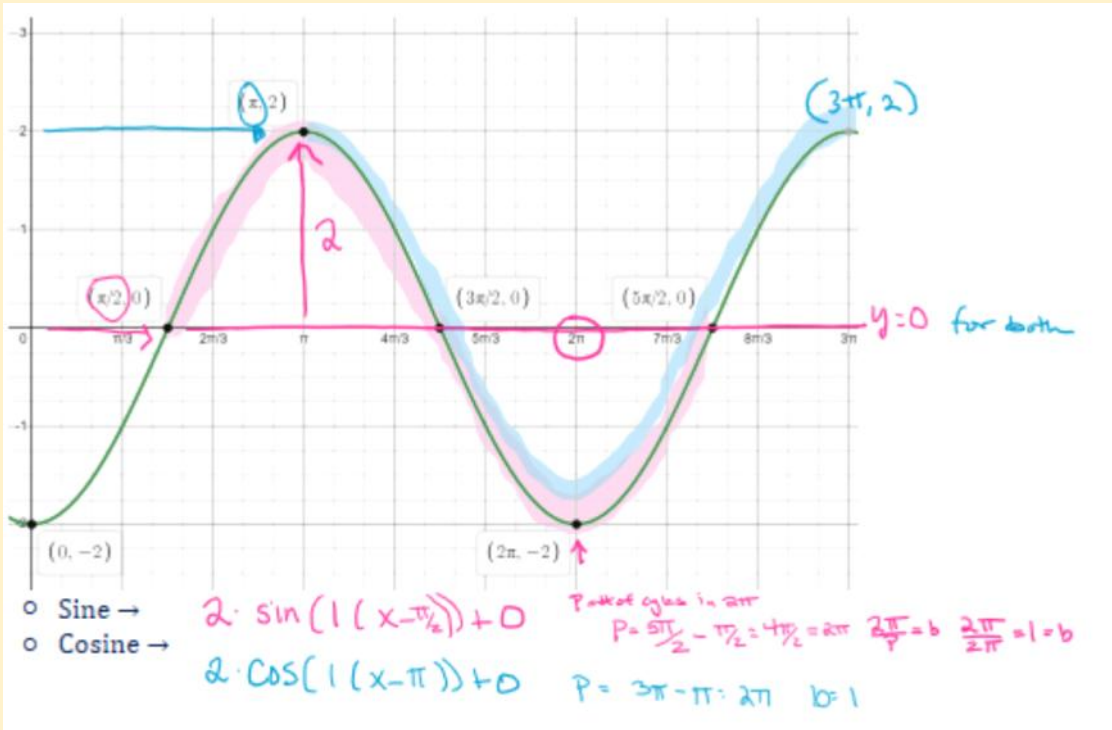
Note,
80 = March 21
172 = June 21
265 = Sept 22
355 = Dec 21

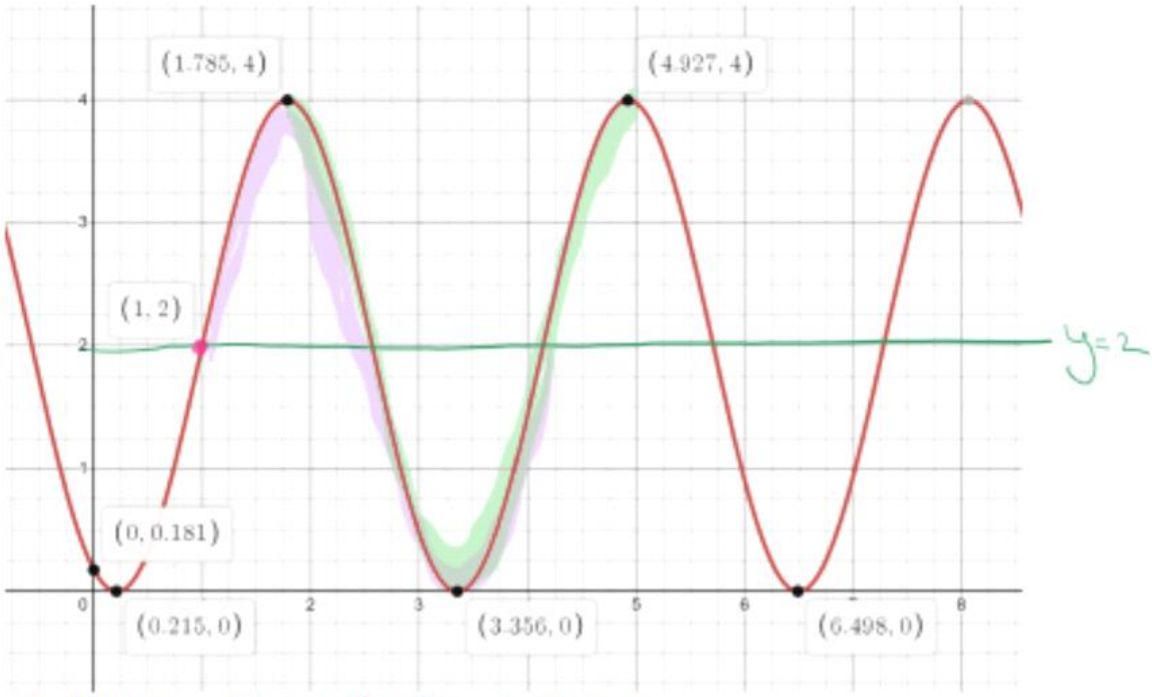
b) Write equations. MAKE SURE TO JUSTIFY HOW YOU KNOW

Sine	*Cosine*
Amplitude, a:	Amplitude, a:
Midline, k:	Midline, k:
Horizontal Translation (Phase Shift), h:	Horizontal Translation (Phase Shift), h:
Period, p:	Period, p:
Frequency, b:	Frequency, b:
Equation:	Equation:

c) Use cosine equation to predict the number of hours of daylight on Veteran's Day (11/11).

Solutions: Remember these are just answers to check your work with. You must show your reasoning to receive full credit.

Equations from Graphs #1:  #2:	2 Problems



- Sine → $2 \cdot \sin(2(x-1))+2$
- Cosine → $2 \cdot \cos(2(x-1.785))+2$ $P=\pi$ $b=2$

or

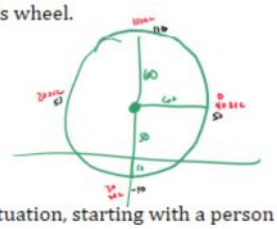
$-2 \cdot \cos(2(x-0.215))+2$

Applications #3	2 Probl

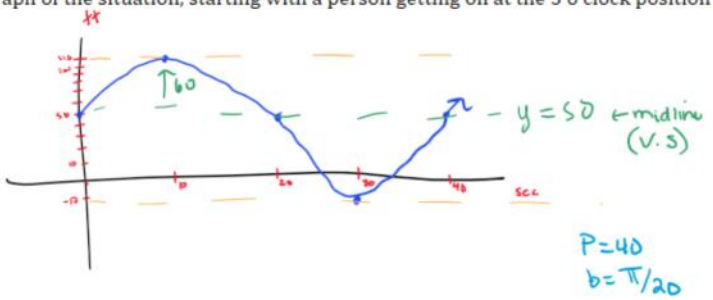
A Ferris wheel 120 feet in diameter completes a revolution, rotating counterclockwise, every 40 seconds. The lowest point is 10 fee below ground.

$r = 60$

a. Draw and label the Ferris wheel.



b. Draw the graph of the situation, starting with a person getting on at the 3 o clock position at time x = 0 seconds.



$P = 40$

$b = \pi/20$

c. Determine an equation to represent the rider's height above the ground. Be sure to keep your equations in radians.

$a = 60$

$P = 40$

$b = \pi/20$

$k = 50$

$$y = 60 \sin\left(\tfrac{\pi}{20}(x)\right) + 50$$

$$\tfrac{2\pi}{40} = \pi/20 = b$$

or

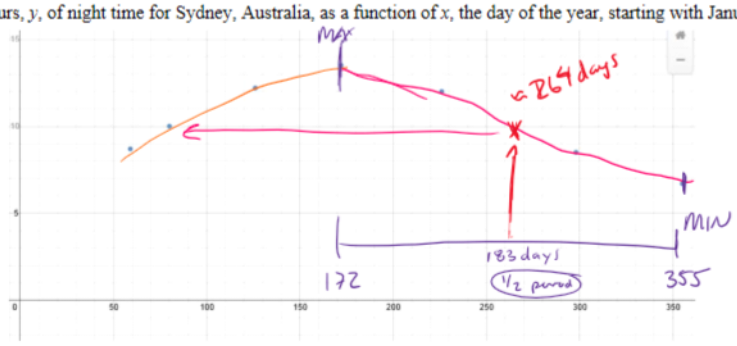
$$y = 60 \cos\left(\tfrac{\pi}{20}(x - 10)\right) + 50$$

#4

ems

The following data gives the number of hours, y, of night time for Sydney, Australia, as a function of x, the day of the year, starting with January 1 as 1.

	x	y
	59	8.7
	80	10
	126	12.2
MAX	172	13.5
	226	12
	265	9.8
	298	8.5
MIN	355	6.7



Note,
80 = March 21
172 = June 21
265 = Sept 22
355 = Dec 21

a) Sketch the graph to fit the data points.
b) Write an equation to fit this data using i) sine and ii) cosine.

MAX = 13.5 hrs.
MIN = 6.7 hrs.

$a = \frac{MAX - MIN}{2} = \frac{13.5 - 6.7}{2} = \boxed{3.4}$

$b = 2\pi/p = 2\pi/366 = \boxed{\pi/183}$

$p = 366$ days

$h =$

$K = MAX - a = \boxed{10.1}$

$$\boxed{y = 3.4 \cdot \sin\left(\frac{\pi}{183}(x - 81)\right) + 10.1}$$

$$\boxed{y = 3.4 \cos\left(\frac{\pi}{183}(x - 172)\right) + 10.1}$$

c) Use cosine equation to predict the number of hours of daylight on Veteran's Day (11/11).

355 = 12/21
335 = 12/1
334 = 11/30
315 = 11/11

plug in 315.